# Mario Becerra-Contreras

**Applied mathematician – Data Scientist – Computer Scientist**

**Location:** Leuven, Belgium

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My main interests are data visualization, statistical modeling (particularly Bayesian modeling), and market research.

## SKILLS

**Statistics**
Bayesian statistics, Experimental Design, Multivariate statistics, A/B testing, MCMC, variational inference, choice modeling

**Machine Learning and Data Mining**
Supervised learning, artificial neural networks, convolutional neural networks, random forests, recommender systems

**Programming languages and other technologies**
R, Python, SQL, Apache Impala, Apache Hive, MATLAB, C++

**Languages**
Spanish: Native English: Native-like competence and accent French: Intermediate (B1) Dutch: Intermediate (A2)

## EDUCATION

**PhD candidate, KU Leuven, Belgium, (2019 – 2023)**
Coursework on statistics and experimental design. I am mostly dedicated to doing full-time research.

**Computer Science MSc, Instituto Tecnológico Autónomo de México (ITAM), Mexico City (2016 - 2018)**
Relevant coursework: Machine learning, Bayesian generalized linear models, Bayesian multilevel modeling, computational statistics, computer architecture, data product architecture, analysis of algorithms, programming language theory.

**Applied Mathematics BSc, Instituto Tecnológico Autónomo de México (ITAM), Mexico City (2010 - 2015)**
Relevant coursework: Probability theory, multivariate statistics, stochastic processes, Bayesian statistics, machine learning, spatial statistics, generalized linear models, programming, data structures, linear programming, numerical optimization, economics.

## EMPLOYMENT

**Freelance data scientist and programmer**
*October 2020 - Present*
Small freelance projects that include:
• Data visualization.
• Speed-up of slow R code by creating C++ functions on the backend.
• Analysis of Twitter data.
• Choice modeling for market research.

**Graduate researcher at KU Leuven, Belgium (Full time)**
*October 2019 - Present*
Research in statistics and experimental design in the Biostatistics group at MeBioS.
I created the *opdesmixr* R package for computing Bayesian D-optimal and I-optimal designs for choice models involving mixtures of ingredients and with the assumption of a multinomial logit model. The package uses C++ under the hood to maximize efficiency. Link to package: https://github.com/mariobecerra/opdesmixr.
I also published a paper with my supervisor, Peter Goos, called *Bayesian I-optimal designs for choice experiments with mixtures* using our package. I presented my research in three international conferences so far.
Modeling of preferences and willingness-to-pay for match tickets of supporters of the Belgian national football teams through statistical analysis of a choice experiment.

**Data scientist at Banco Azteca, Mexico City (Full time)**
*September 2018 - September 2019*
Data analysis for the marketing and CRM teams of the bank.
I was also in charge of creating and optimizing software for these analyses. Additionally, I managed and helped junior analysts in the area. I also presented results to non-technical stakeholders, such as directors and managers of the bank.
Projects included:
• Visualization and analysis of large customer datasets.
• Models for the prediction of future bank transactions.
• Spatial analysis of customer and branch location data.
• Marketing campaign analysis and causal inference.
• Optimized the sampler of a Bayesian hierarchical model for BTYD analysis suited for large scale data analysis.
• Creation of marketing campaigns based on customer analysis, particularly:
  - Debit customer segmentation based on regularity of deposits and withdrawals.
  - Identification of highly regular remittance receivers from the US and Mexico.

## EMPLOYMENT

- Prediction model to identify remittance customers who are willing to accept a bank loan.

**Data scientist at Business Data Evolution, Mexico City (Full time)**
*September 2017 - September 2018*
Developed models both for prediction and for inferential analysis in data science consulting company.
- Analysis of predictive models using Individual Conditional Expectation (ICE) and variable interaction quantification in R.
- Deployed final predictive models in Microsoft Azure VM and Azure Machine Learning.
- Developed Bayesian predictive model for small black and white documents belonging to 4 classes.
- Developed a Bayesian time series demand prediction model for a national grocery store.
- Met with clients to understand their problems, and choose the right model for the data and the clients' needs.
- Presented results to non-technical audiences.

**Data scientist at CAD Salud, Mexico City (Full time)**
*May 2016 - June 2017*
Built predictive and inferential statistical models for health projects.
- Performed statistical analysis, exploratory data analysis and developed models in R.
- Created multivariate predictive model for potential diabetes patients in public hospitals.
- Generated spatial mathematical model for vaccines demand in public hospitals in Mexico.
- Aided in the creation of multivariate index to rank public hospitals.

**Data Scientist at Grupo Salinas, Mexico City (Full time)**
*October 2015 - May 2016*
Designed and built statistical and machine learning models for predictive and inferential analysis in market research.
- Performed statistical analysis, exploratory data analysis and developed models in R.
- Created data visualization dashboards using RStudio's Shiny.
- Aided in creation of predictive model of loan origination clients in a Mexican bank.

**Data analyst at Computer Research and Analysis Laboratory, ITAM, Mexico City (Part time)**
*June 2014 - June 2015*
Natural Language Processing for automatic news classification for media monitoring company.
- Cleansed news articles data for modeling, mainly using R.
- Created different predictive models for automatic news classification in Python using Scikit Learn.

## HOBBIES AND INTERESTS

- **Running**: at least once a week for a few kilometers each time
- **Skiing**: I enjoy this activity very much, even though I am not the best skier out there
- **Hiking**: going to the forest or mountain with friends or colleagues
- **Cycling**: I like long-distance road cycling with friends and team members, but I also sometimes do mountain biking
- **Bicycle maintenance and mechanics**: I like fixing my bikes myself and keeping them in good shape
- **Language learning**: I learned French mostly on my own, and I am currently taking Dutch lessons
- **Reading**: mostly non-fiction titles, but also classic and modern novels
- **Traveling**: I like traveling to different countries and cities to experience new cultures
- **Playing guitar**: I have been playing for 18 years, and I am taking lessons to improve my technique on classical guitar
- **Cooking**: being away from my country made me appreciate and learn the recipes of my culture
- **Generative art**: I started learning how to combine programming and data visualization to create visually appealing images